

ANGEL FIRE, NM, USA

WMO: 720411

Lat: 36.422N

Lon: 105.290W

Elev: 8380

StdP: 10.76

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 00137

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -14.5	(c) -7.0	(d) -19.3	(e) 2.6	(f) -8.1	(g) -12.3	(h) 3.9	(i) 1.0	(j) 34.1	(k) 31.6	(l) 30.4	(m) 32.5	(n) 0.4	(o) 160	(p) 0.717

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 31.6	(c) 79.3	(d) 52.5	(e) 77.2	(f) 52.3	(g) 74.9	(h) 51.8	(i) 57.9	(j) 69.6	(k) 56.9	(l) 68.6	(m) 55.8	(n) 67.6	(o) 11.4	(p) 260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.2	85.7	61.3	52.9	81.8	60.5	51.8	78.6	59.9	29.1	69.9	28.2	68.8	27.4	67.3	61.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 28.4	(b) 24.9	(c) 21.0	(e) -28.1	(f) 83.2	(g) 5.1	(h) 1.9	(i) -31.8	(j) 84.6	(k) -34.7	(l) 85.8	(m) -37.6	(n) 86.9	(o) -41.3	(p) 88.3		
(4) 28.4	(5) 24.9	(6) 21.0	(7) -28.4	(8) 60.3	(9) 5.0	(10) 0.9	(11) -32.0	(12) 60.9	(13) -34.9	(14) 61.5	(15) -37.7	(16) 62.0	(17) -41.4	(18) 62.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.0	20.4	24.7	33.1	39.0	45.4	54.7	58.9	57.0	51.6	41.2	30.7	22.4		
	DBStd	14.91	9.35	9.51	8.14	6.35	5.70	4.10	2.69	3.06	4.22	6.76	8.18	9.71		
	HDD50	4397	918	709	524	332	161	8	0	0	29	281	579	856		
	HDD65	9123	1383	1129	989	781	607	308	189	248	402	737	1029	1321		
	CDD50	749	0	0	0	1	19	150	276	217	78	8	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	668	0	0	0	0	0	262	240	126	40	0	0	0		
	CDH80	37	0	0	0	0	0	25	10	2	0	0	0	0		
	WSAvg	7.2	6.9	9.2	9.0	9.7	8.3	6.9	4.6	4.6	5.1	6.9	6.8	8.0		
	PrecAvg	19.3	0.9	1.1	1.4	1.5	1.5	1.4	3.2	3.0	1.7	1.4	1.0	1.1		
(15) Precipitation	PrecMax	25.5	2.2	1.9	3.0	4.3	4.2	3.2	6.9	6.0	3.3	3.6	2.0	2.5		
	PrecMin	14.3	0.2	0.2	0.3	0.2	0.0	0.0	1.3	1.1	0.3	0.1	0.1	0.2		
	PrecStd	3.0	0.6	0.6	0.7	1.1	1.2	0.8	1.4	1.1	0.9	0.8	0.6	0.7		
	PrecStd	3.0	0.6	0.6	0.7	1.1	1.2	0.8	1.4	1.1	0.9	0.8	0.6	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.0	52.5	62.8	64.6	73.0	82.9	81.4	80.6	78.4	69.9	60.6	50.8		
		MCWB	34.2	36.3	40.8	42.4	46.4	51.6	53.5	52.6	51.1	46.0	40.1	34.3		
	2%	DB	43.8	49.0	58.7	62.7	70.3	80.0	79.1	77.3	74.6	66.0	56.4	46.3		
		MCWB	32.1	35.1	39.3	41.6	45.4	51.1	53.9	53.1	50.5	44.7	38.6	33.0		
	5%	DB	40.8	45.4	54.3	59.4	67.8	77.1	77.0	74.9	72.1	63.0	53.2	42.6		
		MCWB	30.7	33.7	37.1	40.3	44.8	49.9	54.0	53.2	49.7	42.9	37.5	31.7		
	10%	DB	37.0	41.4	50.2	56.6	64.1	74.4	74.0	72.5	69.0	60.2	48.3	38.7		
		MCWB	29.5	31.1	35.3	39.2	43.9	49.3	54.3	53.3	49.8	42.5	35.2	30.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.5	38.4	42.0	45.5	51.2	57.6	59.4	59.0	56.4	52.2	42.5	37.2		
		MCDB	43.5	49.6	58.5	60.6	63.9	70.7	71.5	70.6	66.3	61.0	55.1	44.9		
	2%	WB	34.2	36.0	40.0	43.4	48.8	55.5	58.2	57.6	54.3	48.5	40.4	34.7		
		MCDB	40.9	46.4	56.0	58.2	62.3	69.8	70.2	68.7	66.0	59.3	52.7	42.7		
	5%	WB	31.9	33.9	37.8	41.6	46.7	53.9	57.3	56.6	53.0	45.9	38.2	32.7		
		MCDB	38.5	43.9	52.4	56.3	61.6	68.3	69.1	67.9	64.9	56.6	49.9	40.2		
	10%	WB	30.0	32.1	35.9	39.9	45.1	52.3	56.3	55.4	51.5	43.8	36.3	31.0		
		MCDB	36.6	41.1	48.5	54.9	61.1	67.2	68.1	67.0	63.2	56.5	47.6	38.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	30.2	25.8	27.7	28.3	31.2	37.3	31.6	32.1	33.6	32.1	30.6	26.6		
		MCDBR	35.3	29.9	35.4	35.4	38.9	43.2	37.4	37.2	40.2	39.4	38.1	31.1		
		MCWBR	24.8	17.9	18.2	18.2	18.8	19.4	16.1	16.7	19.0	20.7	22.1	19.9		
		MCDBR	29.3	26.4	31.6	32.8	33.9	35.0	28.9	29.3	31.8	31.7	32.9	26.2		
	5% WB	MCWBR	21.4	16.3	16.7	17.3	17.7	17.4	13.4	14.3	16.3	18.2	19.6	17.7		
(40) Clear-Sky Solar Irradiance	taub		0.222	0.228	0.257	0.277	0.286	0.303	0.342	0.320	0.281	0.264	0.233	0.228		
	taud		2.667	2.662	2.569	2.524	2.555	2.494	2.459	2.490	2.611	2.621	2.656	2.641		
	Ebn at Noon		322	330	324	320	316	308	296	301	311	310	314	312		
	Edh at Noon		21	24	29	33	32	34	35	33	28	25	21	20		
(44) All-Sky Solar Radiation	RadAvg		995	1292	1690	2055	2282	2377	2068	1944	1767	1429	1079	870		
	RadStd		60	92	127	118	158	131	102	97	98	95	63	46		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	+0.54	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+112	+47

Nomenclature: See separate page